Smaran Vallabhaneni

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Education

Olin College of Engineering — Needham, MA
Bachelor of Science, Engineering in Computing (GPA: 4.0) — *Aug. 2025 – May 2029*

- **Relevant Courses:** Linear Algebra, Multi-Variable Calculus, Differential Equations, Modeling & Simulation, Bayesian Statistics, Artificial Intelligence, Machine Learning, Software Design (Python), Kernel Development (C), Algorithms and Data Structures, Cryptography, Data Science, Statistics for Data Science, Data Analysis with Python, Data Preparation & Applied Machine Learning, Predictive Business Analytics, Products & Markets, Principles of Integrated Engineering, Introduction to Sensors, Instrumentation & Measurement

Skills

Programming Languages: C, C++, Rust, Java, Python, R, MATLAB, SQL, TypeScript, JavaScript, C#, Swift, Dart, Bash, LaTeX
Frameworks: PyTorch, Scikit-Learn, LLMs, RAG Systems, Ollama, OpenCV, NumPy, pandas, MediaPipe, ROS, MAVROS, React, Flutter, SwiftUI, Django, AngularJS, Kivy
Tools: GitHub Copilot, Google Gemini, Google Antigravity, Claude Code, LM Studio, Git, IntelliJ, Eclipse IDE, VSCode, Xcode, Node.js, Next.js, Docker, ArduPilot, Figma, Adobe Creative Suite
Operating Systems: Windows, Unix, macOS, Linux (Raspberry Pi OS, Debian, Fedora, and Arch based)
Other: HuggingFace, NLP, Amazon Web Services, Google Cloud, Microsoft Azure, DevOps, API Integration, Testing and Debugging, Continuous Integration Pipelines, UI, UX
Soft Skills: Communication, Problem Solving, Team Collaboration, Leadership, Adaptability, Time Management, Attention to Detail

Projects

W.A.G.E | *Python, TypeScript, Next.js, FastAPI, MySQL, Ollama* — Jul. 2026

- Architected a live job-board feed NLP inference pipeline; postings are tokenized, chunked, and scored via nomic-embed-text embeddings and chatgpt-oss-20b semantic similarity, with sub-5-second end-to-end latency across 25+ concurrent documents.
- Backed by a MySQL-stored FastAPI REST layer for privacy-preserving text extraction; optional Ollama backend swaps cloud inference for local execution with identical query semantics.

Specter | *Python, Linux, Ollama* — May 2026

- Designed a real-time vision-language inference pipeline on Linux/Wayland; routes live screen state to a local gemma4 vision model, achieving 3-second action latency and 86% task classification accuracy across diverse UI environments.

3D Gesture Tracking | *Python, OpenCV, MediaPipe, Blender* — Apr. 2026

- Implemented a real-time hand landmark detection and gesture classification pipeline via MediaPipe and OpenCV; achieves sub-50ms end-to-end latency and 95% precision (5% FPR) across varied lighting conditions and hand geometries.

Whoa-Scope | *Python, Kivy* — Feb. 2026 – Apr. 2026

- Profiled and restructured heap allocation patterns in a real-time signal visualization tool, reducing peak memory consumption by 200+ MB; stabilized operation on memory-constrained lab hardware and drove organic adoption across 40 concurrent users.

ClassicTunes | *Swift, SwiftUI, Metal API, SQLite* — Oct. 2025 – Jul. 2026

- Engineered a native macOS GPU rendering pipeline via Metal API shaders and command encoders (120 FPS CoverFlow on 150+ albums); implemented iPod syncing via reverse-engineered binary protocol with custom bit manipulation schema and AES-128 encryption over a SQLite device-state schema.

Experience

CLOS (Command Line Operating System) — Remote
Founding Engineer and Principal Architect — *Apr. 2023 – Present*

- Founded CLOS, an offline-first mathematical computing platform with ∼1,500 weekly active users; supports graphing, symbolic calculus, simulation, logic design, and AI tutoring across Windows, macOS, and Linux.
- Engineered 8 core subsystems in Java from scratch: custom DB engine, symbolic math parser, scripting language, 3D renderer, terminal, matrix library, logic gate simulator, and symbolic calculus processor; achieves 60 FPS rendering and sub-50ms latency on multi-variable calculus.

Olin Field Robotics Lab — Needham, MA
Robotics Software Engineer — *Sep. 2026 – Present*

- Overhauling the ROS2 software stack for the GroundFleet soil sampling rover, a farm-ng Amiga platform with an onboard Intel NUC coordinating drive control, robotic arm actuation, and onboard soil analysis.
- Developing a semantic segmentation pipeline for real-time crop and field boundary detection, enabling the rover to autonomously identify and avoid active crop rows during field operations.

Olin College AERO Project Team — Needham, MA
Software Engineering Lead — *Jul. 2026 – Present*

- Leading a 6-engineer software sub-team across the full autonomy stack (image recognition, stitching, flight planning, autopilot, and ground station via ArduPilot, ROS, MAVROS, NVIDIA Jetson) for AUVSI sUAS; achieved sub-100ms round-trip telemetry latency and stable 20 FPS at 10-mile field range.
- Drive bi-weekly cross-functional integration cycles with mechanical and electrical sub-teams across 15–20 engineers to synchronize airframe iterations with software requirements.

Activities

Olin ICPC — Needham, MA
Competitive Programmer — *Feb. 2026 – Present*

- Solving high-difficulty algorithmic problems under strict time constraints using segment trees, Fenwick trees, DP, graph algorithms (network flow, SSSP, MST), string algorithms, and computational geometry.

Certifications & Publications

Meta Database Engineer Certification | *Credential ID: M8WKYG9PMT4G* — Jul. 2025

C++ Memory Safety and Integrity Evaluation | *DOI: 10.58445/rars.2876* — Aug. 2025

Performance Evaluation of C++ and Java for Portable Devices | *DOI: 10.58445/rars.1783* — Oct. 2024